

Varnit Sahu

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EDUCATION

University of Waterloo

Bachelor of Computer Science • Artificial Intelligence Specialization • Statistics Minor

- President's Scholarship of Distinction & **Dean's Honours List** (2024 - Present)

Chosen Coursework: Data Structures, Algorithms, Object Oriented Programming, Databases, Hardware, Algebra, Probability

Expected: Aug 2028

Waterloo, ON

EXPERIENCE

PlayStation (Sony Interactive Entertainment)

Sept 2026 – Dec 2026

Software Engineering Intern

San Mateo, CA

- Joining the PlayStation Store & Checkout team supporting and optimizing digital commerce and purchasing experiences for **129 million+** monthly active users using Java, Spring Boot, and AWS

Nokia

Jan 2026 – Apr 2026

Software Developer (Co-op)

Ottawa, ON

- Cut customer provisioning time **from 2 weeks to 3 hours** by automating Terraform-driven cloud infrastructure on GCP
- Spearheaded the development of an agent-to-agent orchestration framework **adopted by 14+ teams** on Kubernetes using ArgoWorkflows to automate artifact versioning, failure diagnosis and perform root-cause analysis across CI pipelines
- Improved cloud-native deployment of LLM recommendation services in Python, reducing Helm/GitOps incidents by **60%**
- Authored scripts to transform REST microservices into MCP tools empowering agentic flows in **3+ years** of legacy services

University Of Waterloo

Dec 2025 – Present

Undergraduate Researcher

Part-Time (Remote)

- Built validation frameworks for Deep Learning to compare PyTorch vs ONNX performance for ML model deployment
- Extended QuackIR , an IR toolkit with SQLAlchemy support for relational databases (PostgreSQL/MySQL)
- Built C++/Linux utilities to automate experiment setup, improving reproducibility of deep learning benchmark studies

Nokia

Sept 2025 – Dec 2025

AI/ML Engineering Intern

Ottawa, ON

- Owned distributed Kubeflow/MLflow pipelines **helping 20+ engineers** automate experiment tracking and model lineage
- Built a RAG model with guardrails over historical incident data, saving **5+ hours weekly** in threat analysis
- Designed and implemented smoke and acceptance test suites using Playwright, improving release reliability by **80%**
- Architected a closed-loop control plane for Autonomous Networks in Azure using python, ingesting real-time telemetry and triggering automated healing workflows, supporting **10K+** concurrent network operations

Savi Finance

Sept 2025 – Dec 2025

Software Engineer Intern

Part-Time (Remote)

- Shipped internal GraphQL microservices on AWS with autoscaling and service routing to **1k+ weekly active users**
- Empowering youth plan long-term investment strategies by modelling age-risks and tax return **saving \$50 in lost fees**
- Built multi-layered caching API for GitLab Runners, to **speed up builds by 60%** (Python, Linux, Docker)
- **Reduced AI inference cost by 25%** by enforcing token budgets, caching prompts, and batching inference requests

WAT.ai

Jan 2025 – Aug 2025

Machine Learning Engineer

Waterloo, ON

- Embedded multi-series line graphs into user dashboards with TypeScript to visualize cost dependencies in related resources
- Built a C++/Kafka-based pipeline to stream real-time training data into Machine Learning model training workflows
- Applied unsupervised learning (K-Means & SINDy) to cluster drilling-regime embeddings and detect operational state shifts using NumPy, Pandas and scikit-learn, reducing anomaly-detection false positives by 27%

PROJECTS

DataBridge (Hackathon Winner) — SQL, Snowflake, MCP, Gemini, Python, React, Javascript

- Led a **team of 4** to architect a **Git-like, versioned SQL pipeline on Snowflake** automating Mergers & Acquisitions
- Implemented a **scalable execution layer** deploying **13+ concurrent workers** to ingest **SQL** metadata, infer schemas, semantically rank/visualize column mappings, and execute deterministic merge operations or create **Jira** tickets on fallback

NetVigil (Advanced Network Security) — Go, Docker, NodeJS, Prometheus, Grafana, Axios

- Developed a rate-limiting reverse proxy that monitors, filters, and protects distributed web applications in real time, providing DDoS mitigation and zero-day threat protection on live traffic
- Delivered a npm-installable package providing Grafana dashboards for enterprise observability with Redis & Prometheus

SKILLS

- **Languages:** C/C++, Python, Golang, Java, Rust, SQL, JavaScript/TypeScript, HTML/CSS, Bash
- **Frameworks/Libraries:** React, NodeJS, Flask, GraphQL, Tailwind, gRPC, NumPy, TensorFlow, PyTorch
- **Tools:** Linux, Git, Docker, Kubernetes, Jupyter, GCP, AWS, CI/CD, PostgreSQL, Redis, Datadog, Codex, Claude Code

Volunteering: Event Organizer at HackCanada 26, NextGEN Student Empowerment Ambassador at Nokia

Interests: Enjoys sports, travelling, cardistry, personal investing, market valuations, strategy games (poker, chess)